

IOC Assessment: 8.8.8.8

Generated 2026-09-01 21:34 UTC Investigation ID: ad8ea627-281d-42d3-81e0-be7282ab48b4

Type: ipv4 **Final Verdict:** benign **Risk Score:** 0.0 (confidence 45.5)

Executive Summary

8.8.8.8 is Google LLC's public DNS resolver (dns.google), a well-known and heavily used legitimate infrastructure address. All three providers that returned reputation data (Spamhaus, VirusTotal, AbuseIPDB) agree on a clean verdict, with a 0/91 detection ratio across VirusTotal engines. The IP is directly allocated to Google LLC (ASN 15169) and is whitelisted by AbuseIPDB. Despite a large volume of historical abuse reports (196 reports from 89 users) — typical of any public DNS service that is frequently targeted — the reputation remains clean. The deterministic scoring engine assigns a malicious probability of 0.0 and an overall risk score of 0.0, consistent with a benign verdict.

Technical Summary

8.8.8.8 is a DIRECT ALLOCATION network block (8.8.8.0/24, NET-8-8-8-0-2) registered to Google LLC (handle GOGL), with network status 'active'. VirusTotal reports a 0/91 detection ratio across all engines, a positive reputation score of 558, and ownership by Google LLC (ASN 15169) in the US. AbuseIPDB classifies the IP as a Content Delivery Network, whitelisted (is_whitelisted: true), with hostname dns.google, and an abuse confidence score of 0. Spamhaus ZEN lookup returns not-listed. The correlation engine confirms the ASN relationship (ipv4:8.8.8.8 --belongs_to_asn--> asn:15169) via VirusTotal. Internet Intelligence surfaced 5 GitHub references, but all relate to unrelated software repositories (SDR drivers, test/automation repos) with no threat connection. Coverage gaps exist: ThreatFox, Censys, and URLhaus were not configured this run, so their data is unknown rather than clean.

Threat Assessment

The evidence strongly supports a benign verdict for 8.8.8.8. Three providers — spamhaus, virustotal, and abuseipdb — all returned a clean reputation with high confidence (Spamhaus and VirusTotal at high confidence, AbuseIPDB at medium). VirusTotal's 0/91 detection ratio across all engines is the strongest single signal, corroborated by a positive reputation score of 558. The IP is unambiguously owned by Google LLC (ASN 15169) and is Google's public DNS service (dns.google), which is legitimate, globally critical infrastructure. The only potentially confusing signals are the 196 abuse reports on AbuseIPDB (from 89 distinct users, most recent 2026-09-01) describing SSH brute force, port scanning, and UFW blocks. These are expected noise against a public DNS resolver that is constantly probed by scanners and attackers; AbuseIPDB itself maintains the whitelist classification and a 0 abuse confidence score, so these reports do not change the clean verdict. VirusTotal also carries a 'suspicious-udp' tag despite the clean verdict, which is a common artifact of DNS traffic patterns rather than evidence of malicious activity. No provider disagrees with the clean verdict. internet_intelligence returned 'unknown' reputation but only because its OSINT collection surfaced unrelated GitHub repositories with no threat connection — this is a data-quality limitation, not a disagreement. The unavailable providers (threatfox, censys, urlhaus) represent coverage gaps, not clean findings, and reduce overall confidence in the assessment.

Verdict Rationale

The verdict is benign, consistent with the given malicious_probability of 0.0. All three providers that returned reputation data (Spamhaus, VirusTotal, AbuseIPDB) agree on a clean verdict, with a 0/91 detection ratio across VirusTotal engines and a positive reputation score of 558. The IP is unambiguously Google LLC's public DNS resolver (dns.google, ASN 15169), directly allocated and active per WHOIS/RDAP, and whitelisted by AbuseIPDB. The 196 abuse reports and 'suspicious-udp' tag are expected noise against a public DNS service and do not change the clean reputation. No provider disagrees with the clean verdict. The overall risk score of 0.0 and severity of 'none' further support a benign classification.

Supporting Evidence

- VirusTotal: 0/91 detection ratio across all engines; IP belongs to Google LLC (ASN 15169) in the US on 8.8.8.0/24; positive reputation score of 558.
- Spamhaus: ZEN lookup returned clean; 'listed' field is false, IP is NOT on the Spamhaus ZEN blocklist.
- AbuseIPDB: reputation score 0, abuse confidence score 0, whitelisted (is_whitelisted: true), classified as Content Delivery Network, hostname dns.google, owned by Google LLC.
- WHOIS/RDAP: 8.8.8.8 is part of /24 network (8.8.8.0-8.8.8.255) DIRECT ALLOCATION to Google LLC (handle GOGL), status 'active'.
- Correlation engine: ipv4:8.8.8.8 --belongs_to_asn--> asn:15169 [virustotal]; provider agreement on 'clean': spamhaus, virustotal, abuseipdb.
- AbuseIPDB: 196 total reports from 89 distinct users (most recent 2026-09-01) including SSH brute force, port scanning, UFW blocks — but these do not change the clean verdict; not flagged as Tor exit node.
- internet_intelligence: 5 GitHub results reference unrelated repos (RTL2832U/R828D SDR driver, SAST test repos) with no connection to the IOC.
- Coverage gap: threatfox, censys, urlhaus were not configured this run (no API key) — their data is unknown, not clean.

Recommended Actions

- No blocking or alerting action is warranted for 8.8.8.8 itself; it is legitimate Google public DNS infrastructure.
- Do not add 8.8.8.8 to blocklists or deny lists; doing so would disrupt legitimate DNS resolution for environments that rely on Google's public resolver.
- If 8.8.8.8 appears as a source in your logs, treat it as expected DNS infrastructure traffic rather than a malicious indicator, unless it is part of a broader suspicious pattern.
- Consider configuring the unavailable providers (ThreatFox, Censys, URLhaus) for future runs to close coverage gaps on this IOC type.

Investigation Priorities

- If 8.8.8.8 is observed in outbound DNS traffic, verify it is consistent with legitimate resolver usage (e.g., configured DNS servers) rather than a sign of compromise.

- Investigate any SSH brute-force or port-scan events attributed to 8.8.8.8 as likely spoofed source addresses or scanner noise, not as evidence the IP is malicious.
- Review whether any internal systems are configured to use 8.8.8.8 as a resolver, as this may be a policy/compliance consideration rather than a security finding.

Incident Response Recommendations

- No incident response actions are required for this IOC given its benign, well-known legitimate nature.
- If an alert fires on 8.8.8.8, validate the surrounding context (protocol, port, associated hosts) before escalating, since the IP is a high-traffic public DNS service.
- Document 8.8.8.8 as an allowlisted/known-good infrastructure address in detection tooling to reduce alert fatigue from expected DNS traffic.